Residual-Shifted Diffusion Models for Efficient Cell Painting Synthesis from Brightfield

Reproducing and Extending CellResDM with Consistency Distillation

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Introduction & Motivation

High-content phenotypic screening, especially using the Cell Painting assay, has driven transformative progress in early-stage drug discovery. This approach enables large-scale, systematic profiling of cellular responses to thousands of perturbations, delivering rich biological insights for both discovery and mechanistic research [1, 2]. Immunofluorescence (IF) imaging remains central to this paradigm, providing molecularly specific visualization of subcellular compartments such as the nucleus, ER, mitochondria, and RNA. However, IF imaging is encumbered by practical constraints: expensive dyes, lengthy and labor-intensive imaging protocols, and the risk of phototoxicity and staining artifacts that can perturb biological measurements [3].

There is therefore growing demand for scalable, label-free imaging alternatives. Bright-field (BF) microscopy provides a compelling substitute by capturing structural information through transmitted light without the need for labelling, resulting in faster, less expensive, and non-destructive imaging [7, 10]. Yet, BF lacks the molecular specificity required for high-content cellular profiling, limiting its utility for detailed biological analyses.

To bridge this gap, recent computational advances have focused on predicting IF images directly from label-free BF images [6, 5]. By virtually converting BF to IF, researchers can harness the cost-effectiveness of BF while recovering much of the phenotypic specificity of IF. However, prior computational approaches each have significant drawbacks: convolutional neural networks tend to generate blurred outputs, while generative adversarial networks can hallucinate artifacts and suffer from unstable training [3, 6, 10]. Modern diffusion models have set a new standard for image quality, but their lengthy inference time limit their use for large-scale screening [5, 4].

CellResDM, a recently introduced residual-shifted diffusion model [11], is a significant advance in this field. It synthesizes high-quality Cell Painting IF images from BF inputs far more efficiently than standard diffusion models—requiring only 15 inference steps while maintaining fidelity to both pixel-level detail and biologically meaningful cell morphology. Building on this foundation, our work further extends CellResDM by integrating consistency distillation, which both accelerates inference and improves reconstruction quality. This positions our improved CellResDM as a practical and scalable tool for routine, high-throughput label-free cell profiling.

Methodology

Our work began with the rigorous reproduction and validation of the CellResDM framework. CellResDM is a residual-shifted diffusion model that leverages the structural similarity between bright-field and immunofluorescence images for efficient synthesis. Rather than generating the IF image from pure noise, CellResDM begins with a noised BF image, with the diffusion process focused on removing noise and correcting the residual difference between BF and IF. At each step, the model predicts the target IF image given the current noised-sample, conditioned on the original BF input and the diffusion timestep. Then, blends this prediction with the evolving noised-sample. This targeted refinement approach dramatically reduces task complexity and enables CellResDM to accurately synthesize IF images in just 15 inference steps.

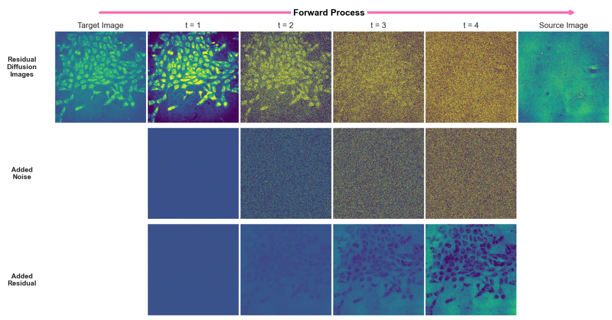

Figure 1: Illustration of the forward process in the residual diffusion framework, shown here for mitochondrial (target) and bright-field (source) images over 4 diffusion steps. The arrows indicate the progression of the forward process, where each step involves adding both noise and a fraction of the residual (BF – IF) to the target image. The reverse (inference) process, not shown, predicts the target image at each step by progressively de-noising and correcting the residual using the current noised image. **Top:** Residual diffusion trajectory at each step. **Middle:** Total noise added at each diffusion step. **Bottom:** Total residuals incorporated at each step.

We then prove that CellResDM fits naturally within the consistency model paradigm [9], a principled framework for drastically reducing the number of inference steps in diffusion-based image synthesis. By explicitly enforcing the model's predictions to be consistent across different diffusion timesteps, our Consistency Distilled CellResDM compresses the inference trajectory from 15 steps down to just two, without sacrificing image quality or biological fidelity. This step-agnostic consistency not only accelerates inference but also provides additional robustness to the model's predictions, improving both pixel-level and cell-level morphological metrics.

For evaluation, we combined standard image-level metrics (MSE, PSNR, SSIM) with a comprehensive suite of cell level morphological analyses. Using CellProfiler, we extracted high-dimensional, single-cell feature vectors from both real and synthetic images, enabling us to quantify the biological relevance of each model's outputs. To achieve a rigorous, cell-level comparison, we performed explicit cell matching between real and synthetic images, which enabled detailed feature correlation analysis. Furthermore, we visualized the global distribution of these features using Uniform Manifold Approximation and Projection (UMAP), which allowed us to assess whether the synthetic cell populations faithfully preserved the diversity and structure of real biological samples—a key consideration for downstream scientific applications.

Key Findings

Performance of CellResDM: CellResDM proved highly effective at synthesizing realistic, biologically meaningful IF images, significantly outperforming widely used baselines like U-Net and Pix2Pix, and achieving competitive results relative to methods with architectures specifically designed for BF-to-IF translation, such as DeepHCS [8]. Quantitatively, CellResDM achieved lower MSE and higher SSIM/PSNR scores, indicating superior reconstruction fidelity. Morphological feature correlations with ground truth were consistently high, confirming that the method preserves detailed cellular structure and fine phenotypic distinctions essential for scientific use.

Impact of Consistency Distillation: By incorporating consistency distillation, our CellResDM variant further advanced —reducing inference from 15 to just 2 steps while simultaneously improving all metrics. This approach achieved the lowest MSE and highest SSIM/PSNR, as well as the strongest feature-level correlations. Crucially, these gains did not come at the expense of increased artifacts or reduced morphological diversity; instead, they enhanced the model's robustness and practical scalability, leading to improved generalization on out-of-distribution data. This improvement can be attributed to the nature of consistency distillation: since the model is no longer directly trained on potentially imperfect ground-truth, distillation acts as an effective regularizer-it prevents the model from over-fitting to artifacts or noise present in the supervised ground truth, thereby promoting more reliable and generalizable representations—particularly valuable in medical imaging settings where annotation and staining artifacts are common.

Limitations on Compared Methods: While DeepHCS performed well on in-distribution data, it failed to generalize, suffering performance drops on out-of-distribution sets. U-Net consistently produced overly smooth, low-detail images, while Pix2Pix introduced frequent artifacts and failed to capture meaningful cell structure. Qualitative inspection and UMAP visualizations confirmed that only CellResDM and its consistency-distilled extension maintained both single-cell detail and the overall distribution of real biological features, whereas other models distorted or collapsed the underlying phenotypic structure.

Conclusion

We recommend Consistency Distilled CellResDM as a robust and efficient solution for scalable, high-throughput, and biologically accurate prediction of IF images from bright-field microscopy. Future research should aim for even greater morphological realism—especially in modeling complex cellular structures—through improved architectures and loss functions. Broader evaluation across diverse cell types and experimental conditions will further validate its generalizability. In summary, our work reproduces and extends CellResDM, establishing it as a state-of-the-art framework for rapid, label-free cell profiling in biomedical research.

References

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